Vulnerability of urban street networks to floods

A case study in São Paulo, Brazil

2025-12-05

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1 Introduction

Objectives

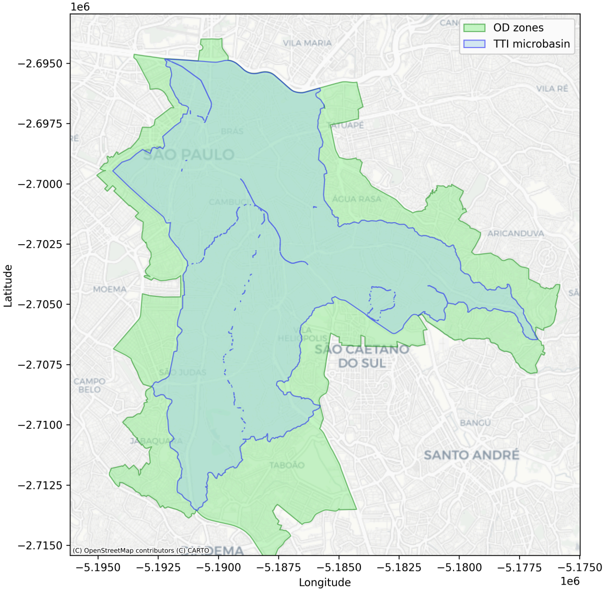
- Investigate the vulnerability of urban street network based on topology and the occurrence of floods.
- Explore the relationship between rainfall, river stage, and flood occurrence, as well as the relationship between topological properties and flood occurrence in the studied area.
- Identify the most vulnerable street sections.

Context

- Resilience of street networks: its capacity to resist, absorb, adapt and transform in the face of environmental impacts (Sharifi, 2019).
- Studying and analyzing the vulnerability of street networks helps in prioritizing planning, budgeting, and maintenance, as well as preparing effective emergency response plans (Balijepalli & Oppong, 2014).

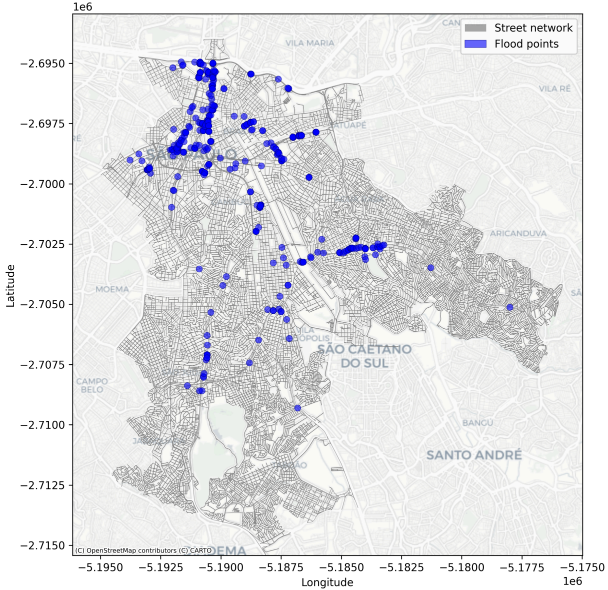
2 Material and methods

Study area and data collection stations



- Tamanduateí (TTI) river basin inside the city of São Paulo
- Origin/destination (OD) zones that intersected the TTI river microbasins

Flood and network data



- Street network from 2025-11 (Boeing, 2025)
- Flood points between 2022-01 and 2025-04

Graph topology

Calculation of:

- Node degree

$$k_i = \sum_j a_{ij}$$

- Edge betweenness centrality (EBC) (vulnerability proxy)

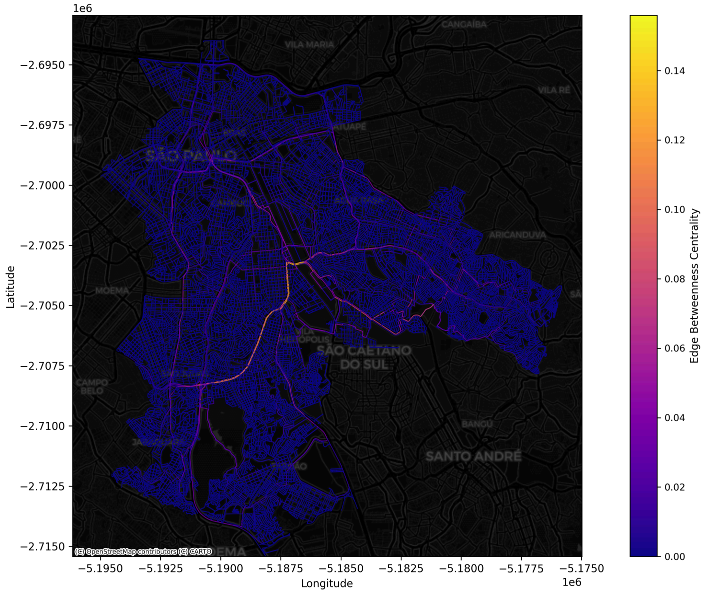
$$c_{B(e)} = \sum_{s,t \in V} \frac{\sigma(s,t \mid e)}{\sigma(s,t)}$$

Floods x Vulnerability

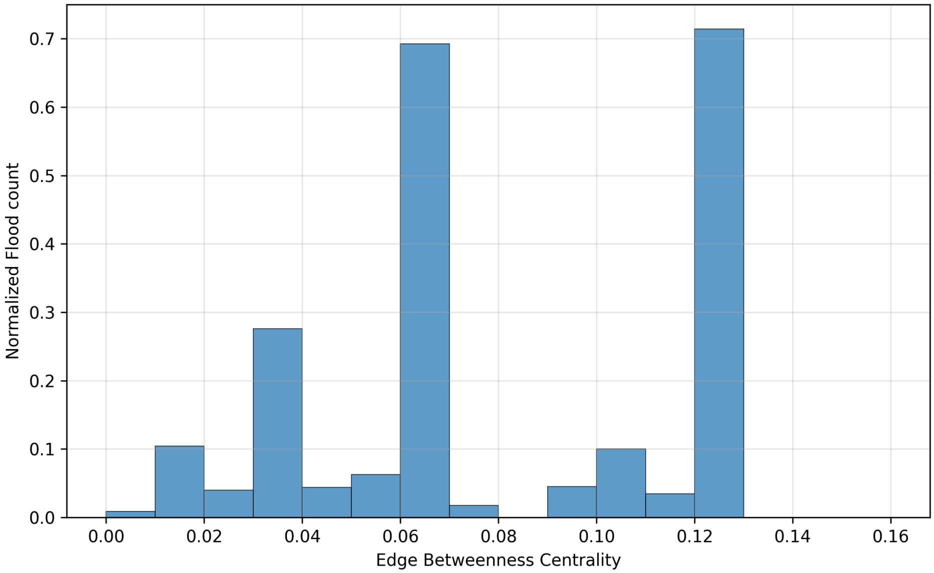
- Comparison between EBC values and flood counts per edge (spatial proximity join)
- Highlight of higher values cases - values above the median, for EBC and flood counts values

3 Preliminary results

Edge betweenness centrality



EBC x Floods


0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Normalized Flood count
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
Edge Betweenness Centrality

4 Bibliography

Bibliography

Balijepalli, C., & Oppong, O. (2014). Measuring Vulnerability of Road Network Considering the Extent of Serviceability of Critical Road Links in Urban Areas. *Journal of Transport Geography*, *39*, 145–155. https://doi.org/10.1016/j.jtrangeo.2014.06.025

Boeing, G. (2025). *Modeling and Analyzing Urban Networks and Amenities with OSMnx*. https://doi.org/10.48550/ARXIV.2505.00736

Sharifi, A. (2019). Resilient Urban Forms: A Review of Literature on Streets and Street Networks. *Building and Environment*, *147*, 171–187. https://doi.org/10.1016/j.buildenv.2018.09.040